DP19. Rigid-body joints projected along the constraint gradient taken at the start of the step (SHAKE, the constrained Verlet step) inside augmented vertex block descent and small-step XPBD, with exact free flight and a splitting rotation as the inertial target, so that swinging jointed bodies keep their amplitude

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Classification (CPC): A63F 13/57 (simulating properties, behaviour or motion of objects in the game world); G06T 13/20 (3D animation); G06F 30/20 (design optimisation, verification or simulation); G06F 17/13 (differential equations); G06F 2111/10 (numerical modelling); B25J 9/1605 (simulation of manipulator lay-out, design, modelling of manipulator); G16C 10/00 (computational theoretical chemistry, molecular mechanics and molecular dynamics).

Related disclosures in this series: DP3 (exact integer time: the fixed clock of 1/480 s steps that the solvers here run on).

Derived means computed exactly by the arithmetic described here, in double precision (IEEE 754 binary64), on a point mass on a massless rigid rod, against the exact pendulum by Jacobi elliptic functions. "One possible choice" marks a detail filled in to make the method buildable, where others work too.

1. Field

Numerical simulation of rigid bodies connected by joints: a time-stepping rule for hard joints inside two fast iterative solvers, which keeps the energy of a swing instead of damping it.

2. Problem and known methods

A hard joint (a hinge, a ball joint, a rod, a weld) must hold at the end of every step. Fast solvers enforce it by moving a predicted pose back onto the joint. **The direction of that move decides whether swings keep their energy.**

Method	What it does	What it lacks for swinging joints
Störmer–Verlet [1, 6]	Positions advanced with the forces at the step's start; second order, symmetric, symplectic	No constraints
SHAKE [2], RATTLE [3]	Verlet with each constraint force along its gradient **at the step's start**, scaled so the constraint holds at the step's end; RATTLE adds velocities. Symplectic [4]. Standard in molecular dynamics	Fails when a constraint's direction turns a lot within a step [5, 9]. Not used inside block descent or XPBD

Method	What it does	What it lacks for swinging joints
Implicit constraint direction [9]	Gradient at the step's end, chosen over SHAKE's for robustness at large cloth steps	Dissipative, like implicit Euler
XPBD [10], small steps [11], rigid bodies [12]	Projects the prediction along the gradient **at the current iterate**; "a close approximation of implicit Euler" [12]	Damps swings
Orientation in XPBD [15]	Lie-group rotations and constraint gradients for rigid bodies and rods	Keeps XPBD's projection at the current iterate
VBD [13], AVBD [14]	Implicit Euler as a minimisation, body by body over a colouring; AVBD adds hard constraints by an augmented Lagrangian with ramped stiffness	Converges to implicit Euler; its authors name the backward Euler step's energy loss and leave BDF2 as future work [14]. Its rotational target turns at constant angular velocity
Free rigid-body splitting [7]; rigid-body SHAKE-type methods [8]	Exact-momentum rotation; symplectic constraints on rigid molecules	Not used as the target of block descent or XPBD

The damping. Both fast solvers, converged on a hard joint, are implicit Euler in the joint's angle. For a pendulum of natural angular frequency ω, at leading order:

$\theta_{n+1} - 2\theta_n + \theta_{n-1} = -(\omega h)^2 \sin\theta_{n+1}$ (gradient at the current iterate)

so the amplitude decays as $\exp(-\omega^2 hT/2)$: a damping ratio $\omega h/2$. A 30° swing at 1/480 s steps keeps **67.1%** of its amplitude after 10 s at 1 Hz and **20.0%** at 2 Hz, in both solvers (derived, §6). Substeps shrink the loss only in proportion. As published, AVBD (1/60 s steps) keeps 3.7% and 0.0%, and XPBD (20 substeps of 1/60 s) 84.8% and 51.8% (derived, linear theory). Doors, pendulums, chains and hanging loads slow down when they should not. A BDF2 projection keeps the amplitude (99.993%) but drifts in phase (0.615 mm off the exact path at 1 Hz).

3. Summary

Each hard joint row's gradient is evaluated once at the step's start and frozen; the joint pushes only along these directions, while the joint function is evaluated at the current iterate, so the joint holds exactly at the step's end. In augmented vertex block descent each joint's force becomes $-(k\cdot C(x) + \lambda)\cdot\partial C(x^t)/\partial x$, whose fixed point is the SHAKE step; in small-step XPBD each joint correction moves the bodies along the frozen gradient with SHAKE's Newton denominator. The free-motion target is made exact: the stored velocity is read as v(t − h/2), so the plain update follows the exact parabola under a constant force, and orientation advances by a symmetric splitting of the free rigid body, which keeps angular momentum exactly. Together these turn both solvers' joints from implicit Euler, which damps swings, into the constrained Störmer–Verlet step, which is second order and symplectic: a 30° swing keeps 100.000% of its amplitude over 10 s and stays within 0.078 mm (1 Hz at 1/480 s) and 0.039 mm (2 Hz at 1/960 s) of the exact path (derived).

4. Detailed description

4.1 Notation and data

Symbol	Meaning
h	The step: for example 1/480 s; smaller for islands with fast swings (§4.9)
x, q	A body's centre of mass and orientation (unit quaternion)
v	Stored linear velocity, **read as v(t − h/2)**
L	Stored world angular momentum, read as L(t − h/2); it replaces a stored angular velocity
M, W	Per body: mass m and world inertia $I_w = R(q)\cdot I_{body}\cdot R(q)^T$; $W = M^{-1}$
C_j	One scalar joint row (a ball joint has three); zero when the joint holds
G^0_j	The row's gradient with respect to both bodies' positions and small world rotations θ, **at the step's start**

Symbol	Meaning
λ_j, k_j	AVBD's multiplier and stiffness; XPBD's accumulated multiplier λ_j
$\tilde{\alpha}_j$	XPBD's compliance divided by h^2 (zero for a hard joint)

Per joint row (one possible choice): G^0 as four 3-vectors (linear and angular parts for bodies a and b), λ, k, and optionally C at the step's start: 13 numbers, 104 bytes in f64 or 52 in f32. A ball joint's three rows can share one compact form: their linear parts are ±the unit axes, so only the two lever arms at the step's start are stored.

4.2 Why the time of the gradient matters

Verlet's step is a kick and a drift: $v(t + h/2) = v(t - h/2) + h \cdot a(t)$, then $x(t + h) = x(t) + h \cdot v(t + h/2)$. The kick uses forces at time t, centred between the two half-step velocities, so the step is symmetric and second order.

A hard joint adds a reaction $-W \cdot G^T \lambda$. With G taken at the step's start, the kick stays centred on t (SHAKE). With G taken at the new pose, where current-iterate projection converges, the joint's push is a whole step late (Figure 1). The leading-order equation becomes

$\theta_{n+1} - 2\theta_n + \theta_{n-1} = -(\omega h)^2 \sin \theta_n$ (gradient at the step's start)

which has no decay. Its frequency is off by about $(\omega h)^2/24$, so the path error grows linearly with time: about $g \cdot \theta_0 \cdot \omega \cdot h^2 \cdot T/24$ (0.058 mm for 1 Hz, 30°, 1/480 s and 10 s by linear theory; the nonlinear run gives 0.078 mm).

Only the direction is frozen. C itself is evaluated at the current iterate, so the joint holds at the step's end.

4.3 The inertial target: exact free flight and exact-momentum rotation

(a) Free flight by the half-step velocity reading. The stored velocity is $v(t - h/2)$. The step is $v \leftarrow v + h \cdot a_{free}(t)$; $y = x + h \cdot v$. Under a constant acceleration this lands on the exact parabola every step: a 5 m/s throw at 60° stays within 6×10^{-11} mm of the exact path over 2 s at 1/480 s (round-off). Read as $v = v(t)$, the same update errs by $\frac{1}{2} \cdot g \cdot h \cdot t$: 20.4 mm after 2 s (derived). The arithmetic equals the published $y = x + h \cdot v + h^2 \cdot a$; only the meaning of v changes, so velocities are converted where they enter or leave:

Boundary	Conversion (one possible choice)
In: a throw, a release, an outside impulse at time t	Store $v(t) - (h/2) \cdot a(t)$, with a the free acceleration; for a jointed body only its part along the joint's free directions matters (the projection removes the rest)
Out: display, sound, a contact's speed, a hand's grip	$v(t) = \frac{1}{2} \cdot (v(t - h/2) + v(t + h/2))$, exact in free flight; for a jointed body optionally projected onto the free directions (RATTLE's velocity step [3])
Angular, with a torque τ	$L(t - h/2) = L(t) - (h/2) \cdot \tau(t)$, read out the same way

(b) Rotation by splitting. The free rigid body's energy, $\frac{1}{2} \cdot \Sigma\, L_k^2 / I_k$ in the body frame, splits into three parts, each an exact rotation about one body axis [7]. Over h, run axis 1 for h/2, axis 2 for h/2, axis 3 for h, axis 2 for h/2, axis 1 for h/2. For each part, with axis k and duration τ:

$\varphi = \tau \cdot L_{body,k} / I_k$; $q \leftarrow q \otimes r_k(\varphi)$; $L_{body} \leftarrow r_k(-\varphi) \cdot L_{body}$, with $r_k(\varphi) = (\cos \varphi/2, \sin \varphi/2 \cdot e_k)$

World $L = R(q) \cdot L_{body}$ stays exactly constant, because each rotation applied to q is undone in L_{body}. The result $\tilde{q}$ is the angular target. Sines and cosines come from the program's own fixed polynomials, so the target is bit-identical across machines where that is needed.

Free spin about the middle axis, 10 s at 1/480 s (box 0.30 × 0.20 × 0.10 m, 1 kg, 10 rad/s; derived)	World L	Energy	Flips (exact: 4)
Constant angular velocity (AVBD's published target)	off 1.2%	exact	0
Explicit gyroscopic term (rigid-body XPBD)	off 4.2%	+8.5%	4
Splitting (this target)	**5×10^{-14}**	**5.4×10^{-6}, bounded, $\propto h^2$**	**4**

4.4 Freezing the joint directions

At the step's start, for each hard joint row j between bodies a and b, compute $G^0_j = \partial C_j/\partial(x_a, \theta_a, x_b, \theta_b)$, with θ a small world-frame rotation, and keep it for the step.

Joint	Rows C_j	Linear parts (a; b)	Angular parts (a; b), lever arms at the step's start
Ball	anchor_a − anchor_b, anchor = x + R(q)·r (3 rows)	I; −I	$-[R(q^a)r_a]\times$; $+[R(q^b)r_b]\times$
Rod of length ℓ	$\lvert d\rvert - \ell$, d = anchor_a − anchor_b	n; −n, n = d/|d| at the step's start	$(R(q^a)r_a) \times n$; $-(R(q^b)r_b) \times n$
Hinge	Ball rows, plus u_a·v_b and u_a·w_b (u_a the axis on a; v_b, w_b across it on b)	0; 0	u_a × v_b; −(u_a × v_b), and the same with w_b
Weld, slider, universal	Built the same way from their rows	—	—

$[r]\times$ is the cross-product matrix. An active limit or motor row is a hard row for that step.

4.5 In augmented vertex block descent

```
step(h):
  for each body i:                                  # the inertial target (4.3)
      x0_i, q0_i = x_i, q_i
      v_i += h * a_free_i ;  y_i = x_i + h * v_i
      L_i += h * tau_i ;     qt_i = split_flow(q_i, L_i, h)
      x_i, q_i = y_i, qt_i                          # or the solver's own initial
guess
  for each hard joint row j:
      G0_j = gradient of C_j at (x0, q0)            # frozen for the step (4.4)
      lambda_j *= r_lambda ;  k_j = max(r_k * k_j, k_start)
  repeat N times:                                   # N = 2-4 at 1/480 s (one
possible choice)
      for each colour, for each body i of that colour (in parallel):
          H = M_i / h^2
          f = -(M_i / h^2) * ((x_i, q_i) minus (y_i, qt_i))
          for each hard joint row j touching i:
              F = k_j * C_j(current poses) + lambda_j
              f -= F * G0_j[i]
              H += k_j * outer(G0_j[i], G0_j[i])
          add contact rows as published
          (x_i, q_i) = (x_i, q_i) plus solve6x6(H, f)
      for each hard joint row j:
          lambda_j = k_j * C_j(current poses) + lambda_j
          k_j += beta * abs(C_j(current poses))
  for each body i:
      v_i = (x_i - x0_i) / h                        # v(t + h/2)
      L_i -= h * sum over rows j at i of lambda_j * G0_j[i].angular
```

"Minus" between poses is AVBD's: the position difference and the rotation vector $2\cdot\mathrm{vec}(q\cdot\tilde{q}^{-1})$. "Plus" applies a rotation $\Delta\theta$ by $q \leftarrow \mathrm{normalise}(q + \tfrac{1}{2}(0, \Delta\theta)\cdot q)$. Against the published loop only the frozen G^0 (in f and H) and the target lines change.

The fixed point. When the sweeps converge (no change and C = 0), each body satisfies $(M/h^2)\cdot(x - y) = -\Sigma_j \lambda_j \cdot G^0{}_j$. With y from §4.3(a) that is the SHAKE step [2, 4]. The published force, along $\partial C(x)/\partial x$, converges instead to implicit Euler. The matrix term $k\cdot G^0 G^{0\intercal}$ is one possible choice: AVBD's own $k\cdot\nabla C(x)\nabla C(x)^\intercal$, or the Newton form $k\cdot G^0\nabla C(x)^\intercal$, changes how fast the sweeps converge, not where. If AVBD's partial error correction is kept (the joint driven to $\alpha\cdot C(x^t)$ rather than 0), the fixed point is the SHAKE step with that correction.

Momentum. Each row applies equal and opposite impulses to its two bodies, so total linear momentum is exact. Total angular momentum changes by $C(x^t) \times$ (the row's impulse): zero when the joint held at the step's start.

4.6 In small-step XPBD

```
substep(h):
  for each body:  x0, q0 = x, q
                  v += h * a_free ;  x += h * v
                  L += h * tau ;     q = split_flow(q, L, h)
  for each joint row j:  G0_j = gradient at (x0, q0) ;  lambda_j = 0
  repeat P times:                                    # P = 1 as in XPBD; more for
coupled hard joints
      for each joint row j:
          c = C_j(current poses)
          d = gradC_j(current poses) . W . G0_j      # SHAKE's Newton denominator
          if d < rho_min * (G0_j . W . G0_j): guard (4.8)
          dlam = -(c + alpha_j / h^2 * lambda_j) / (d + alpha_j / h^2)
          lambda_j += dlam
          for each body b of row j:
              x_b += G0_j[b].linear * dlam / m_b
              rotate q_b by I_w_b^-1 * G0_j[b].angular * dlam
  contacts and the velocity pass: as published
  for each body:  v = (x - x0) / h                     # v(t + h/2)
                  L += sum over rows j of G0_j[b].angular * lambda_j / h
```

For one rod, each correction is a Newton step along the frozen direction, with SHAKE's quadratic convergence. A cheaper variant uses $d = G^0 \cdot W \cdot G^0$ and converges linearly. A soft joint ($\alpha > 0$) keeps XPBD's compliance term; only its direction is frozen.

4.7 Parameters

Parameter	One possible choice	Useful range
Step h	1/480 s; 1/960-1/1920 s for islands with fast swings	1/120-1/3840 s; error $\propto h^2$
AVBD sweeps a step	2-4	1-10
AVBD k_start	About $0.4 \cdot m/h^2$ for the body's mass (10^5 N/m for 1 kg at 1/480 s, §6.2)	0.1-1 × m/h^2
AVBD β	10^7 N/m² for 1 kg at 1/480 s (§6.2), scaled with m/h^2	Tuned by measurement
Warm start r_λ, r_k	AVBD's per-step factors (λ by $\alpha \cdot \gamma$, k by γ; $\alpha = 0.95$, $\gamma = 0.99$, set at 1/60 s) at the same per-second rate: $r_\lambda = (\alpha \cdot \gamma)^{(60h)}$, $r_k = \gamma^{(60h)}$	0.9-1 per 1/60 s
XPBD passes P	1 for single joints; 2-8 for chains	1-20
Guard ρ_min	0.5	0.3-0.8
Residual tolerance	0.1 μm for gameplay bodies	0.01-10 μm

4.8 Safeguards and convergence

When SHAKE has no answer. If a joint's direction turns a lot within one step, or a rod's length changes a lot, no scaling of the step-start direction restores the joint, and the denominator d goes to zero: SHAKE's known failure configurations [5, 9]. The ratio $\rho_j = (\nabla C_j(x) \cdot W \cdot G^0{}_j)/(G^0{}_j \cdot W \cdot G^0{}_j)$ measures it; for one rod it is the cosine of the turn. The guard (one possible choice): if $\rho_j < 0.5$, a turn of more than about 60° in a step, the island's step is split in two; where that is not allowed, the row uses the current-iterate gradient for that step and is flagged. At 1/480 s the 1 Hz swing of §6 turns 0.4° a step. A lever arm spinning at ω turns $\omega \cdot h$, so the guard trips only above 503 rad/s (4,800 rpm). If $|C|$ stays above the tolerance after the sweeps, more sweeps run or the step is split.

Convergence. In XPBD a single row converges quadratically near its answer; coupled rows solved Gauss–Seidel converge linearly, as SHAKE does in molecular dynamics, and long chains need more passes or a direct tree solve. In AVBD the joint force is no longer the gradient of an energy, so the block descent no longer minimises a function. One body with one joint converges to the SHAKE step in 2-4 sweeps (derived, §6.2). Since G^0 and $\nabla C(x)$ differ only by the step's small turn, the change is a small

perturbation of AVBD's variational form. Chains, closed loops, large mass ratios (up to 50,000:1) and contacts in the same island rely on the residual check above: where |C| stays above the tolerance, more sweeps run or the step is split.

4.9 What runs where, and the cost

Where	Precision	Used for
CPU, job workers by island	f64; fixed order of operations; no fused multiply-add differences; own sines and cosines	Bodies that affect gameplay; bit-identical on every machine
GPU compute shaders, vendor-neutral API	f32; positions within 2 m of an island origin	Cosmetic bodies such as debris; bit-identical on one GPU

The method's own cost is small: G^0 is computed once a step (a few dozen operations a row) instead of every sweep, and the splitting adds five small rotations a body a step (about a few hundred operations, less than one 6 × 6 sweep). Fast swings set the step per island: at 30°, the path line of §6 holds over 10 s at 1/480 s up to about 1 Hz, at 1/960 s at least to 3 Hz and at 1/1920 s at least to 5 Hz (the frequencies tested; derived). Only those islands pay for the smaller step.

5. Variants

The method may also:

- **Freeze the gradient at another symmetric time:** RATTLE's form (positions as here, velocities projected at the step's end); higher-order symmetric compositions (three SHAKE steps as a fourth-order triple jump); a split step taking the gradient at the start of each part.
- **Live in other iterative solvers:** VBD, AVBD, PBD, XPBD with or without substeps, sequential impulses with a position projection, projective dynamics, ADMM, Newton or interior-point solvers; Jacobi instead of Gauss–Seidel; blocks of one body, several bodies or a whole island; colouring by bodies or by rows.
- **Cover every joint type:** ball, hinge, slider, universal, weld, rod, rope segment, gear, pulley, closed loops, motors and active limits; inextensible cloth and hair edges; Cosserat rods; hard volume rows of soft bodies; incompressibility rows of particle fluids.
- **Treat soft joints** with frozen directions too, or freeze only hard rows. Resting contacts that act as bilateral rows for a step may also take frozen directions.
- **Use other exact or near-exact targets:** the ballistic law between contacts; the exact free rigid-body flow by elliptic functions; other splittings, axis orders and compositions; Lie-group integrators; rigid-body SHAKE or RATTLE on the rotation matrix [8]. Or freeze joint directions with the published targets alone, which already fixes the swing of point-like bodies.
- **Store state differently:** two positions instead of a velocity (position Verlet); momenta instead of velocities; angular velocity instead of angular momentum, converted at the step's start.
- **Choose other safeguards:** a guard ratio of 0.3-0.8; a line search; relaxation of 0.5-1.5; a post-stabilisation projection; sweep counts set by a residual tolerance.
- **Run at any precision or on any hardware:** f64, f32, mixed (f64 positions about an origin, f32 increments), fixed point for positions; CPU SIMD, GPUs of any maker, FPGAs, many machines by island.
- **Serve other fields:**

Field	Use
Robotics	Arms, legged robots, grippers and cable-driven parts in simulation; model-based control and reinforcement learning, where numerical damping biases what is learned; digital twins
Biomechanics and medicine	Skeletons with joints; gait and sports swings; prosthesis and exoskeleton design; surgical simulators with jointed instruments

Field	Use
Molecular dynamics	Bond and angle constraints, and rigid molecules, inside block-parallel GPU solvers; coarse-grained models
Film and animation	Chains, ropes, hanging props, hair and cloth that keep their energy without hand tuning
Engineering and CAD	Mechanism studies, cranes and suspended loads, pendulum dampers in buildings, vehicle suspensions, deployable booms and tethered satellites
Training and teaching	Virtual-reality simulators and physics teaching tools that must keep energy

6. Worked example

6.1 The swing test

A point bob hangs from a fixed pivot on a massless rigid rod of length $\ell = g/\omega_0^2$ (0.2485 m for 1 Hz, 0.0621 m for 2 Hz; $\omega_0 = 2\pi \cdot f$), is released at 30° and runs 10 s. The exact motion is $\sin(\theta/2) = k \cdot cd(\omega_0 t \mid k^2)$, $k = \sin(\theta_0/2)$. The run starts exactly: $v(-h/2) = (x(0) - x(-h))/h$. **Kept amplitude** is the largest $|\theta|$ in the last 1.5 periods divided by θ_0; **path error** the largest distance between the simulated and exact bob over 10 s. Each step's joint is solved exactly (a closed form for one rod). **Pass lines:** amplitude within 4.76% and path within 0.0925 mm, set from published limits of the best human vision: a 5% speed discrimination divided by 1.05, and a stereo acuity of 5 arcseconds, which resolves about 0.37 mm of depth at 1 m with the eyes 6.5 cm apart and, since depth thresholds grow with the square of distance, 0.0925 mm at a viewing distance of 0.5 m.

Integrator (derived)	Step	1 Hz: kept / path error	2 Hz: kept / path error	Pass lines
Current-iterate gradient, converged (implicit Euler: AVBD and hard XPBD)	1/480 s	67.143% / 59.4 mm	19.990% / 32.3 mm	Fails both
The same, as published (linear theory)	AVBD 1/60 s; XPBD 1/1200 s	3.7%; 84.8%	0.0%; 51.8%	Fail
BDF2 projection	1/480 s	99.993% / 0.615 mm	99.890% / 1.156 mm	Path fails
Step-start gradient (SHAKE)	1/480 s	**100.000% / 0.078 mm**	100.000% / 0.154 mm	1 Hz passes; 2 Hz path fails
Step-start gradient (SHAKE)	1/960 s	100.000% / 0.020 mm	**100.000% / 0.039 mm**	Both pass
Implicit Euler: substeps of 1/480 s needed	—	9 for amplitude, 579 for path	34 / 578	(5 Hz: 211 / 574)

At 1 Hz and 1/480 s the step-start gradient keeps the amplitude within 0.002% from 5° to 90°; the path error grows with the swing: 0.0096 mm at 5°, 0.031 mm at 15°, 0.078 mm at 30°, 0.256 mm at 60°, 0.527 mm at 90° (derived). Large swings therefore take smaller steps (§4.9). Figures 2 and 4 plot these runs.

6.2 One body in a block descent

The loop of §4.5 on one body (1 kg) with one rod joint, from a cold start ($\lambda = 0$), without partial error correction ($\alpha = 1$), warm start γ^⅛ with $\gamma = 0.99$, k_start = 10^5 N/m and $\beta = 10^7$ N/m² at 1/480 s (scaled by $(1/480\text{ s} \div h)^2$ otherwise), initial guess y (derived):

Joint direction in the force	Sweeps	1 Hz at 1/480 s: kept / path error	2 Hz at 1/960 s: kept / path error
Current iterate (published)	2	67.145% / 59.35 mm	43.406% / 29.36 mm
Current iterate (published)	4	67.144% / 59.35 mm	43.405% / 29.36 mm
Step start (this paper)	2	100.003% / 0.072 mm	100.008% / 0.033 mm
Step start (this paper)	4	100.001% / 0.076 mm	100.002% / 0.037 mm

The largest joint error is 26 μm (2 sweeps) and 8.7 μm (4 sweeps) at 1/480 s, in the first step after the

cold start, and under 0.3 μm after the first second. With 32 sweeps the loop reproduces SHAKE's path error (0.0781 mm), and with the published direction implicit Euler's (59.35 mm): the loop's fixed point for one body is the SHAKE step (§4.5).

7. Figures

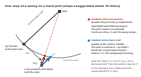


Figure 1. One step of a swing on a hard joint, exaggerated about 75 times. The free-flight prediction y is off the joint's circle. Red: moved along the gradient at the new position, where AVBD and XPBD converge; the push is a step late and the swing loses energy. Blue: moved along the rod's direction at the step's start (SHAKE); the push is centred, as Verlet's forces are, and the swing keeps its energy.



Figure 2. Amplitude of a 30° swing over 10 s at 1/480 s steps (derived). With the current-iterate gradient the swing decays to 67.1% (1 Hz) and 20.0% (2 Hz); with the step-start gradient both keep 100.000%. Dotted: the 4.76% pass line.

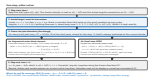


Figure 3. One step in either solver. Shared: the half-step velocity reading, the splitting rotation, the frozen joint directions and momentum from the joints' impulses. Steps 4a and 4b are the two solvers' loops; both converge to the SHAKE step.



Figure 4. Worst path error of a 30° swing over 10 s against the step (derived, log scales). The step-start gradient falls as h^2 and meets the 0.0925 mm line at 1/480 s for 1 Hz and 1/960 s for 2 Hz; the current-iterate gradient stays tens of millimetres off.

8. References

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